

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Machine Learning

Exam: IN2064 / Endterm

Date: Monday 19th February, 2024

Examiner: Prof. Stephan Günnemann

Time: 17:00 – 19:00

	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8	P 9
I									

Working instructions

- This exam consists of **16 pages** with a total of **9 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 38 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - Two-sided DIN A4 sheet of handwritten notes (a print of digitally handwritten notes is allowed).
- **No other material (e.g. books, cell phones, calculators) is allowed!**
- Physically turn off all electronic devices, put them into your bag and close the bag.
- There is scratch paper at the end of the exam (after problem 9).
- Write your answers only in the provided solution boxes or the scratch paper.
- If you solve a task on the scratch paper, clearly reference it in the main solution box.
- All sheets (including scratch paper) have to be returned at the end.
- **Only use a black or a blue pen (no pencils, red or greens pens!)**
- **For problems that say “Justify your answer” you only get points if you provide a valid explanation.**
- **For problems that say “Derive” you only get points if you provide a valid mathematical derivation.**
- **For problems that say “Prove” you only get points if you provide a valid mathematical proof.**
- If a problem does not say “Justify your answer”, “Derive” or “Prove”, it is sufficient to only provide the correct answer.

Left room from _____ to _____ / Early submission at _____

Problem 1 Decision Tree (4.5 credits)

You want to approach a *regression* problem using a decision tree. You are given the subsequent dataset consisting of a single feature and one regression target.

	Feature x	Target y
1	0.1	0
2	0	-1
3	1.3	4
4	0.4	1

The predictions are obtained via averaging the labels of the corresponding leaf nodes. We construct the tree greedily using binary splits. As split criterion the Squared Error gain $g(\mathcal{Y}, \mathcal{Y}_L, \mathcal{Y}_R) = e(\mathcal{Y}) - e(\mathcal{Y}_L) - e(\mathcal{Y}_R)$ is optimized. $\mathcal{Y}$ denotes the set of regression targets y before the split, while $\mathcal{Y}_L, \mathcal{Y}_R$ are the two sets of regression targets after applying a prospective split. The Squared Error is defined as $e(\mathcal{Y}) = \sum_{y_i \in \mathcal{Y}} (y_i - \bar{y})^2$ with prediction $\bar{y}$.

0 ☐ 1 ☐ a) To improve the purity do we need to minimize or maximize the Squared Error gain? **Justify** your answer!

0 ☐ 1 ☐ b) Calculate the predictions for all possible binary splits resulting in depth 1 trees.

c) Which of the possible binary splits in (b) results in the best possible split of the data. **Justify** your answer!

☐	0
☐	1
☐	2

Problem 2 Probabilistic Inference (4 credits)

0

1

2

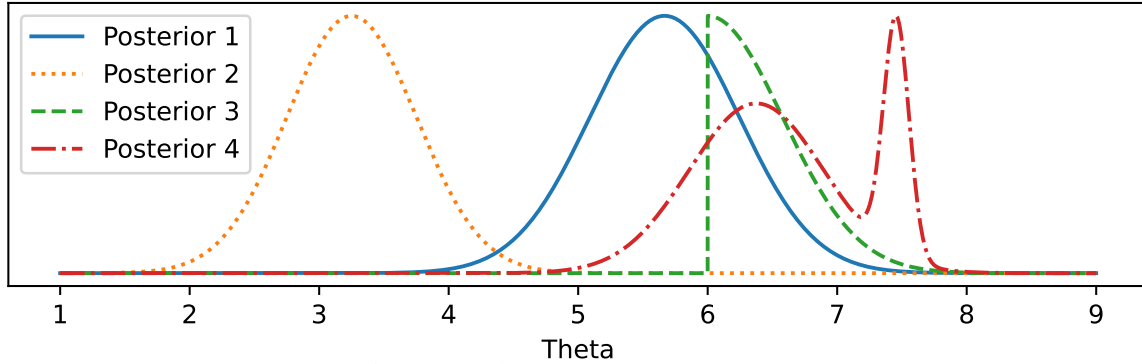
3

4

Consider the probabilistic model

$$p(x | \theta) = \mathcal{N}(x | \theta, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(x - \theta)^2\right)$$

and a set of observations $\{x_1, \dots, x_N\} = \{10, 3, 5\}$ consisting of 3 samples $x_i \in \mathbb{R}$. The following figure shows a plot of different unnormalized **posterior distributions** $p(\theta | \mathbf{x})$ over θ . Your task is to match the prior distributions $p(\theta)$ from (a) to (d) to each of the posteriors 1 – 4.



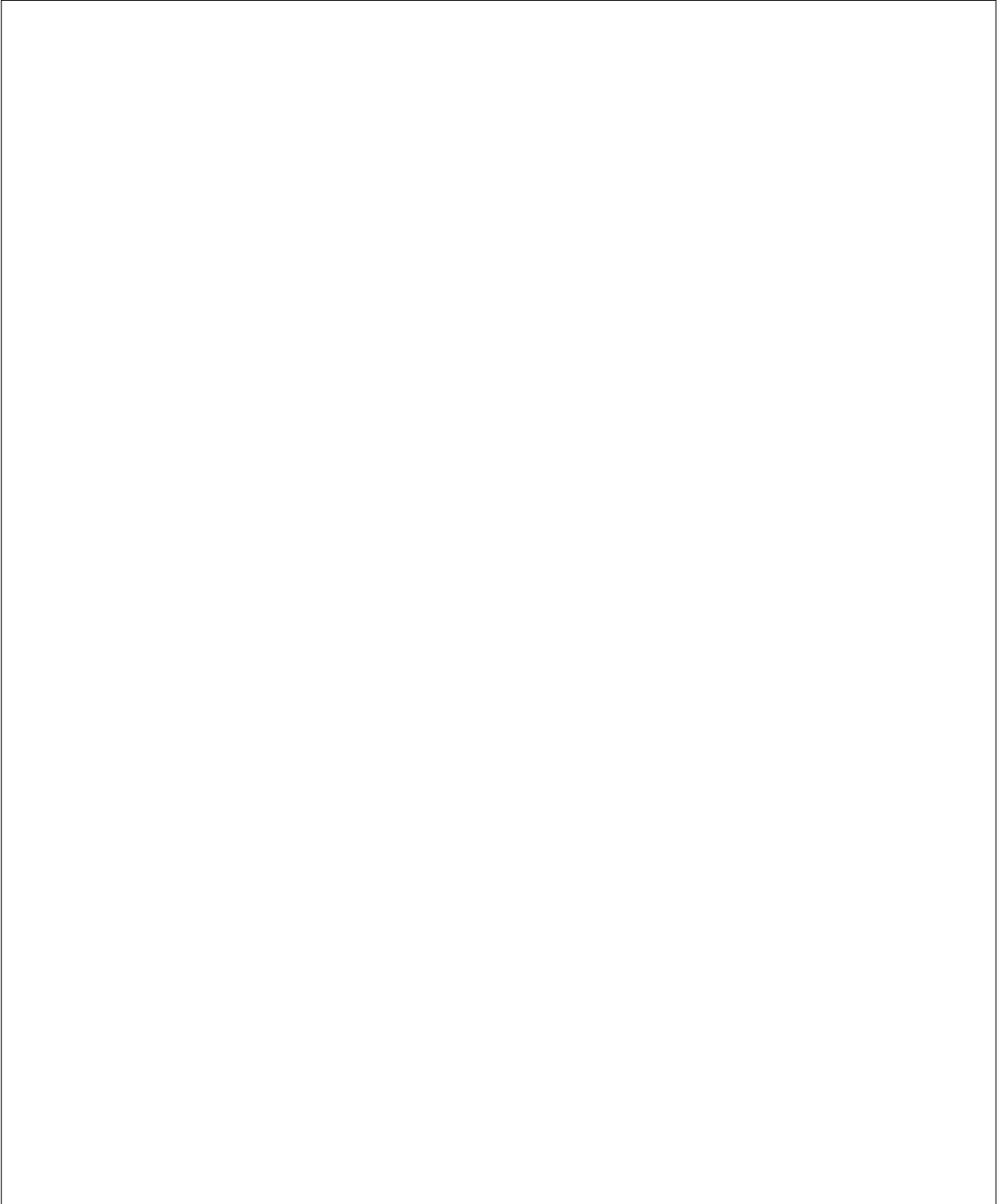
a) $p(\theta) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(\theta + 5)^2\right),$

b) Uniform prior, with $a = 6, b = 8,$

c) $p(\theta) = \frac{1}{2} \exp(-|\theta + 5|),$

d) $p(\theta) = 0.5 \frac{1}{0.1\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\theta - 7.5}{0.1}\right)^2\right) + 0.5 \frac{1}{1.2\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\theta - 8}{1.2}\right)^2\right)$

Which posteriors result from applying priors a) to d). **Justify** your answers.



Problem 3 Linear regression (5 credits)

The goal of linear regression is to find an optimal $\mathbf{w}^*$ to approximate $\mathbf{y} \approx \Phi \mathbf{w}^*$ with the design matrix $\Phi \in \mathbb{R}^{n \times d}$.

0 ☐
1 ☐

a) Could we solve the optimization problem by simply computing $\mathbf{w}^* = \Phi^{-1} \mathbf{y}$? **Justify** your answer.

0 ☐
1 ☐

b) The closed form solution for linear regression is given by $\mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$. Explain why the pseudo inverse $(\Phi^T \Phi)^{-1} \Phi^T$ might not be computable and provide a possible factor in the data that can lead to it. **Justify** your answer.

c) When applying ridge regression the closed-form solution changes to $\mathbf{w}_{ridge}^* = (\Phi^T \Phi + \mu \mathbb{I})^{-1} \Phi^T \mathbf{y}$. **Prove** that if $\mu > 0$ then $(\Phi^T \Phi + \mu \mathbb{I})$ is always invertible.

Hint: A square matrix is invertible iff none of the eigenvalues are zero. Further, any covariance matrix $\Phi^T \Phi$ is positive semi-definite, i.e., for all eigenvalues we know $\lambda \geq 0$.



Problem 4 Linear Classification (3 credits)

Consider a dataset of scalar datapoints $x_i \in \mathbb{R}$ with binary labels $y_i \in \{0, 1\}$. We now apply a polynomial feature map $\phi_d(x) = (1, x, x^2, \dots, x^d)^\top$ to the datapoints and consider a linear classifier:

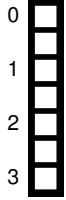
$$p(y_i = 1 | x_i, \mathbf{w}) = \sigma(\mathbf{w}^\top \phi_d(x_i)) = \frac{1}{1 + \exp(-\mathbf{w}^\top \phi_d(x_i))}$$

Here, $\sigma(\cdot)$ is the sigmoid function.

Prove that the decision boundary of this classifier is a polynomial of degree d . That is, show that the decision boundary is of the form

$$\sum_{k=0}^d \alpha_k x^k = 0$$

for some $\alpha_k \in \mathbb{R}$ with $k \in \{0, 1, \dots, d\}$.



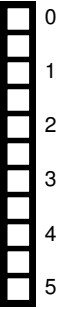
Problem 5 Optimization (5 credits)

Consider a convex quadratic function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as:

$$f(x) = a(x - b)^2 + c \quad \text{where } a > 0$$

We want to minimize $f(x)$ using the gradient descent algorithm with a fixed learning rate τ . For which $\tau > 0$ does this method converge to the minimizer b of $f(x)$? **Prove** your answer.

Hint: The method converges if $f(x^{(t+1)}) < f(x^{(t)})$.



Problem 6 Support Vector Machines and Kernels (3 credits)

0	☐
1	☐
2	☐
3	☐

Given $a \in \mathbb{R}_{>0}$, $b \in \mathbb{R}$, and $\mathbf{x}, \mathbf{y} \in (0, 1)^n$, i.e., $x_i, y_i \in (0, 1) \forall i \in \{0, \dots, n\}$. **Prove** that

$$k(\mathbf{x}, \mathbf{y}) = \frac{a}{1 - \min_i x_i \min_j y_j} + b^2$$

is a kernel.

Hint: $\frac{a}{1-r} = \sum_{k=0}^{\infty} ar^k$ for $|r| < 1$.

You can use the kernel preserving operations from the lecture, given again as reference.

Let $k_1 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and $k_2 : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be kernels, with $\mathcal{X} \subseteq \mathbb{R}^N$. Then the following functions k are kernels as well:

1. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) + k_2(\mathbf{x}_1, \mathbf{x}_2)$
2. $k(\mathbf{x}_1, \mathbf{x}_2) = c \cdot k_1(\mathbf{x}_1, \mathbf{x}_2)$, with $c > 0$
3. $k(\mathbf{x}_1, \mathbf{x}_2) = k_1(\mathbf{x}_1, \mathbf{x}_2) \cdot k_2(\mathbf{x}_1, \mathbf{x}_2)$
4. $k(\mathbf{x}_1, \mathbf{x}_2) = k_3(\phi(\mathbf{x}_1), \phi(\mathbf{x}_2))$, with the kernel k_3 on $\mathcal{X}' \subseteq \mathbb{R}^M$ and $\phi : \mathcal{X} \rightarrow \mathcal{X}'$
5. $k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1^T \mathbf{A} \mathbf{x}_2$, with $\mathbf{A} \in \mathbb{R}^N \times \mathbb{R}^N$ symmetric and positive semidefinite

Problem 7 Deep Learning (3 credits)

Consider the following code snippet representing the implementation of a function $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, where n is the dimension of the input vectors $\mathbf{x}$ and $\mathbf{y}$. Assume $\mathbf{y} \in \mathbb{R}_{>0}^n$.

```
1 class F:
2     def forward(self, x: np.array, y: np.array) -> np.array:
3         self.cache = x, y
4
5         # MISSING CODE FRAGMENT
6
7         assert out.shape == x.shape
8         assert out.shape == y.shape
9         return out
10
11    def backward(self, d_out: np.array) -> (np.array, np.array):
12        x, y = self.cache
13
14        # CODE FRAGMENT TO BE UPDATED
15        # d_x = d_out.copy()
16        # d_y = d_out.copy()
17        # for i in range(len(x)):
18        #     d_x[i] *= -y[i]
19        #     d_y[i] *= 1 / y[i] - x[i]
20
21        assert d_x.shape == x.shape
22        assert d_y.shape == y.shape
23        return d_x, d_y
```

a) **Update** the highlighted code fragment in the backward method of F (lines 15-19) with the respective vectorized implementation.

☐ 0
☐ 1

b) **Write** one possible vectorized implementation of the forward method of F (missing code fragment on line 5). Multiple lines of code are allowed.

☐ 0
☐ 1
☐ 2

Problem 8 Autoencoder / Dimensionality Reduction (6.5 credits)

Consider a dataset of real-valued datapoints $\{\mathbf{x}_i\}_{i=1}^N$ with $\mathbf{x}_i \in \mathbb{R}^D$, i.e., $\mathbf{X} \in \mathbb{R}^{N \times D}$ and $\text{rank}(\mathbf{X}) = r$. We would like to train an autoencoder f with a K -dimensional hidden layer with $K < D$. There are no biases and no activation functions, i.e., $f(\mathbf{x}) = \mathbf{W}_2^T \mathbf{W}_1^T \mathbf{x}$, with $\mathbf{W}_1 \in \mathbb{R}^{D \times K}$ and $\mathbf{W}_2 \in \mathbb{R}^{K \times D}$. f is trained to minimize the following reconstruction error:

$$\mathcal{L} = \sum_{i=1}^N (\mathbf{x}_i - f(\mathbf{x}_i))^T (\mathbf{x}_i - f(\mathbf{x}_i))$$

- 0 ☐ a) Write down the weights $\mathbf{W}_1$ and $\mathbf{W}_2$ of a global minimizer for the three cases $r > K$, $r = K$, and $r < K$.
1 ☐ **Justify** your answers.
2 ☐
3 ☐

b) When will the reconstruction loss be zero and how many optimal solutions exist for each of the three previous cases. **Justify** your answers.

☐	0
☐	1
☐	

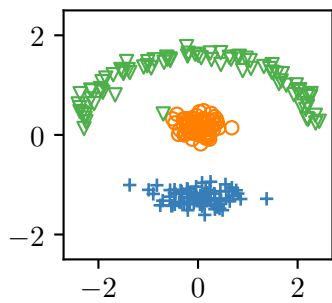
c) Assume $\mathbf{W}_1$ is column orthonormal. How do the pairwise Euclidean distances between the original and projected data points change for the cases $r > K$ and $r = K$? **Justify** your answers.

☐	0
☐	1
☐	2

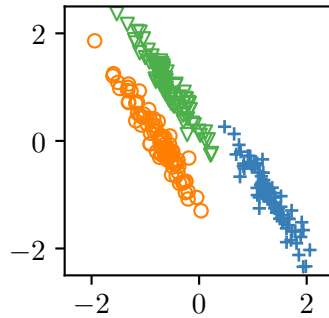
Problem 9 Clustering (4 credits)

0 ☐ 1 ☐ 2 ☐

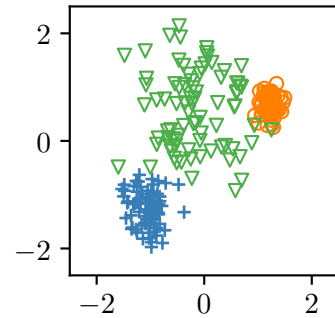
a) Consider the following three two-dimensional datasets. Each figure shows the most likely cluster assignments based on a Gaußian Mixture Model (GMM) after convergence with $K = 3$ number of clusters. Each symbol represents one cluster assignment.



(a)



(b)



(c)

The table below contains the methods used to create the cluster predictions. Assign each Figure a), b), and c) to their respective clustering method. **Justify** your answers.

	Method	Σ_k
(i)	GMM	$\Sigma_k = \sigma_k I$
(ii)	GMM	No constraints on Σ_k
(iii)	GMM	Σ_k diagonal

b) Show that there exists a degenerate case where the proxy objective used in the expectation maximization (EM) algorithm, and thus the likelihood, becomes unbounded when applying the EM-algorithm to GMMs. **Justify** your answer.

Hint: The probability density function of the multivariate Gaussian with mean $\mu \in \mathbb{R}^D$ and covariance matrix $\Sigma \in \mathbb{R}^{D \times D}$ is

$$\mathcal{N}(\mu, \Sigma) = (2\pi)^{-D/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right).$$



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

